



Department of Creative Technologies

Full Stack Development Lab

Semester Project

Muhammad Shamoil (232025)
BSSE-6-A

[GITHUB PROFILE](#)



24/7 Virtual Access

Instant connection to your dedicated care team and elite specialists anytime, anywhere. Experience healthcare without boundaries.



Global Specialist Network

Priority access to the world's leading medical minds, selected for their clinical brilliance and outcomes. Your health, globalized.



Executive Health Assessments

Comprehensive diagnostic screenings and preventative protocols designed for high-performance lifestyles. Optimization through data.



THE CORNERSTONE

The Annual Assessment

Our proprietary assessment protocol goes beyond traditional medicine. We leverage cutting-edge genomics, high-resolution imaging, and functional biomarkers to create your unique biological blueprint.

- Whole-Genome Sequencing
Identifying future risks with predictive precision.
- Advanced Metabolic Profiling
Real-time data on nutrition and hormonal optimization.
- Elite Cardiac Screenings
Non-invasive, AI-enhanced cardiovascular mapping.

Ready to elevate your healthcare journey?

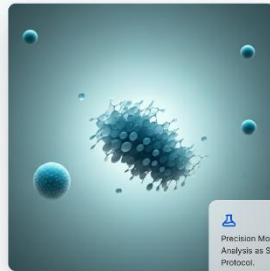
Admission to lifeCore Concierge is limited to maintain our unparalleled standard of individual attention and care precision.

Inquire for Membership

ELITE CONCIERGE

Redefining Medical Excellence

A sanctuary of elite care, built on a foundation of uncompromised precision and exclusive service. We bridge the gap between advanced technology and human-centric empathy.

[Explore Our Philosophy](#)[Request a Private Tour](#)

Precision Molecular Analysis as Standard Protocol.

The lifeCore Difference



Zero Compromise

We reject the status quo of high-volume medicine. Each member receives an exhaustive, individual-first approach where time is never an obstacle to diagnostic depth.



Holistic Precision

Integrating epigenetic mapping, lifestyle data, and specialist consultation into a singular, cohesive health strategy designed for longevity.



99.8%
DIAGNOSTIC ACCURACY



24/7
GLOBAL ACCESS



Top 1%
SPECIALISTS ONLY



Dr. Julian Vance
Chief of Medical Strategy

Former Head of Research at NeoGen Institute.



Dr. Elena Rossi
Director of Clinical Excellence

Recipient of the 2022 Global Innovation Award.



Marcus Thorne
Managing Director

Global Hospitality & Logistics Specialist.

Redefine your baseline

Secure your invitation to join an exclusive network where healthcare is tailored to your unique biological signature.

[Inquire for Membership →](#)

2. Module I: Authentication System

The authentication system is highly secure and role-based, categorizing users into `Admin`, `Doctor`, and `Patient` roles. It prevents unauthorized API access across all protected endpoints.

Implementation Details:

- **Secure Registration:** Users provide their details, which are rigorously validated. Passwords are securely hashed using `bcrypt` before database insertion.
- **OTP Email Verification:** To ensure the validity of user emails, a 6-digit OTP is generated upon registration and sent to the user's inbox via a custom HTML-formatted Nodemailer template. The user must verify this code to unlock their account.
- **JWT Login & Logout:** Upon successful verification and login, a JWT token is generated. Protected routes restrict dashboard access strictly to authenticated sessions. Secure logout functionality is implemented to clear cookies and session data.
- **Role-Based Access Control (RBAC):** Express middleware checks the JWT payload to ensure users can only access endpoints matching their specific roles.

The screenshot displays the 'Join lifeCore Concierge' registration page. At the top, the 'lifeCore' logo is on the left, and a 'Secure Login' button is on the right. The main heading is 'Join lifeCore Concierge', followed by a subtext: 'Complete your registration to access personalized health insights and world-class medical support.' The form is divided into two columns: 'Personal Details' and 'Medical History'. The 'Personal Details' column includes fields for 'Full Name' (filled with 'Patient-Test'), 'Email Address' (filled with 'shamoli003@gmail.com'), 'Password' (masked with dots), 'Age' (filled with '21'), and 'Phone' (filled with '+9276876876'). The 'Medical History' column includes a 'Blood Group' dropdown (filled with 'A+'), a 'Known Allergies' text area (filled with 'Peanuts , Penicillin , ~~Lactin~~ , Lactose'), and a 'Chronic Diseases' section with a list containing 'Synositis' and 'HyperTension'. A blue '+' button is next to 'Synositis', and a red trash icon is next to 'HyperTension'. Below the form is a large black button labeled 'Register as Patient'. At the bottom, there is a link: 'Already have an account? [Login](#)'. A small 'N' logo is in the bottom left corner.

lifeCore

Secure Login

Join lifeCore Concierge

Complete your registration to access personalized health insights and world-class medical support.

Personal Details

Medical History

Full Name

Patient Test

Email Address

shamoi003@gmail.com

Password

Age

21

Phone

+9276876876

HyperTension

2009

Lactose

Verify Your Email

We've sent a 6-digit verification code to shamoi003@gmail.com.

Verify Email

Register as Patient

Already have an account? Login

```
◇ injected env (9) from .env // tip: ◇ secrets for agents [www.dotenvx.com]
Server running on port 5000
MongoDB Connected: 127.0.0.1
Message sent: <40818029-df2c-6d5a-bdd5-7c243ae00c32@gmail.com>
```

Suggested task

Use verification code now • Due on Jun 7

Remind me

By Gemini; there may be mistakes. Learn more

LifeCore Hospital

to me ▾

10:16 PM (0 minutes ago) ☆ 🗑️ ↶ ⋮

lifeCore

CONCIERGE MEDICAL

Verify Your Email Address

Welcome to lifeCore! To complete your registration, please use the verification code below. This code will expire in 10 minutes.

457895

If you did not attempt to register an account with lifeCore, please ignore this email.

© 2026 lifeCore Concierge. All rights reserved.

↶ Reply

↷ Forward

🗑️

Personal Details

Medical History

Full Name

Patient-Test

Email Address

shamoi003@gmail.com

Password

Age

21

Phone

+92768768764

HyperTension

+

✕

Add any long-term medical conditions.

Verify Your Email

We've sent a 6-digit verification code to shamoi003@gmail.com.

457895

Verify Email

lifeCore

Welcome Back

Secure access to your health portal

Email Address

email@example.com

Password

Forgot Password?

Sign In

New to lifeCore?

Register as Patient

Register as Doctor

PREMIUM CARE

lifeCore Concierge

24/7 Virtual Consultations

Direct access to elite specialists anytime, anywhere.

Secure Medical Records

State-of-the-art encryption for your health data privacy.

Personalized Wellness Plans

Data-driven roadmaps tailored to your unique biology.

Join over 10k members receiving elite care

© 2025 lifeCore Concierge. HIPAA Compliant System.

Join our Elite Medical Network

Partner with lifeCore Concierge to provide world-class medical care through our innovative, high-tech patient platform.

Personal Details

Full Name

Dr. Jane Doe

Email Address

jane.doe@truhealth.com

Password

Specialization

Pediatrics

Experience (Years)

10

Qualifications

MD, Ph.D. from Harvard Medical School... (comma separated)

Schedule Availability

Available Days

Mon

Tue

Wed

Thu

Fri

Sat

Sun

Start Time

09:00 AM

End Time

05:00 PM

Day Slot Visualization (1-hour blocks)

PREP

ACTIVE CONSULTATION

BREAK

09:00 AM

10:00 AM

01:00 PM

03:00 PM

05:00 PM

Register as Doctor

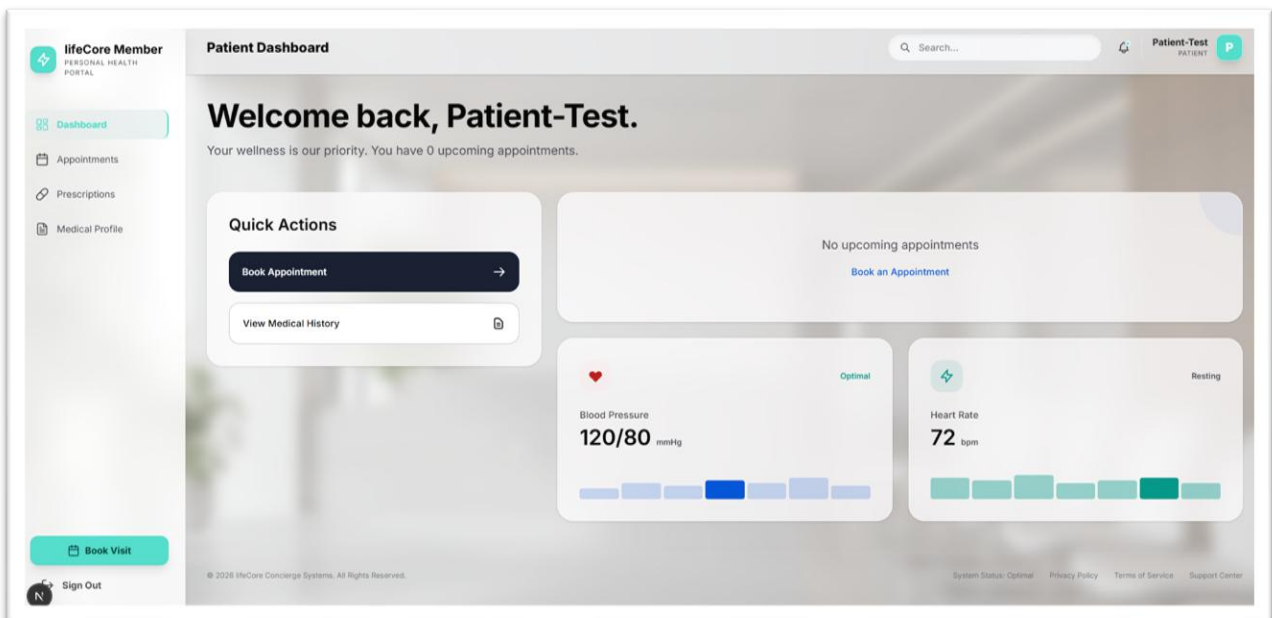
Already have an account? [Login](#)

3. Module II: Doctor & Patient Management System

This module encompasses the core CRUD operations for handling hospital records, strictly adhering to the 15-record data constraint to simulate a fully-populated concierge clinic.

Implementation Details:

- **Data Models:** Separate Mongoose schemas for `User`, `DoctorProfile`, and `PatientProfile` ensure that generic auth data is kept separate from specialized medical data (e.g., allergies, chronic diseases, schedules).
- **CRUD Operations:** Fully functional API endpoints allow administrators and doctors to add, view, update, and delete patient records and doctor profiles.
- **Doctor-Patient Assignment:** The system supports dynamic assignments where doctors are explicitly linked to specific patients for long-term care management.
- **Input Validation:** Both the Next.js frontend forms and Express backend controllers rigorously validate all input fields to maintain database integrity.
-

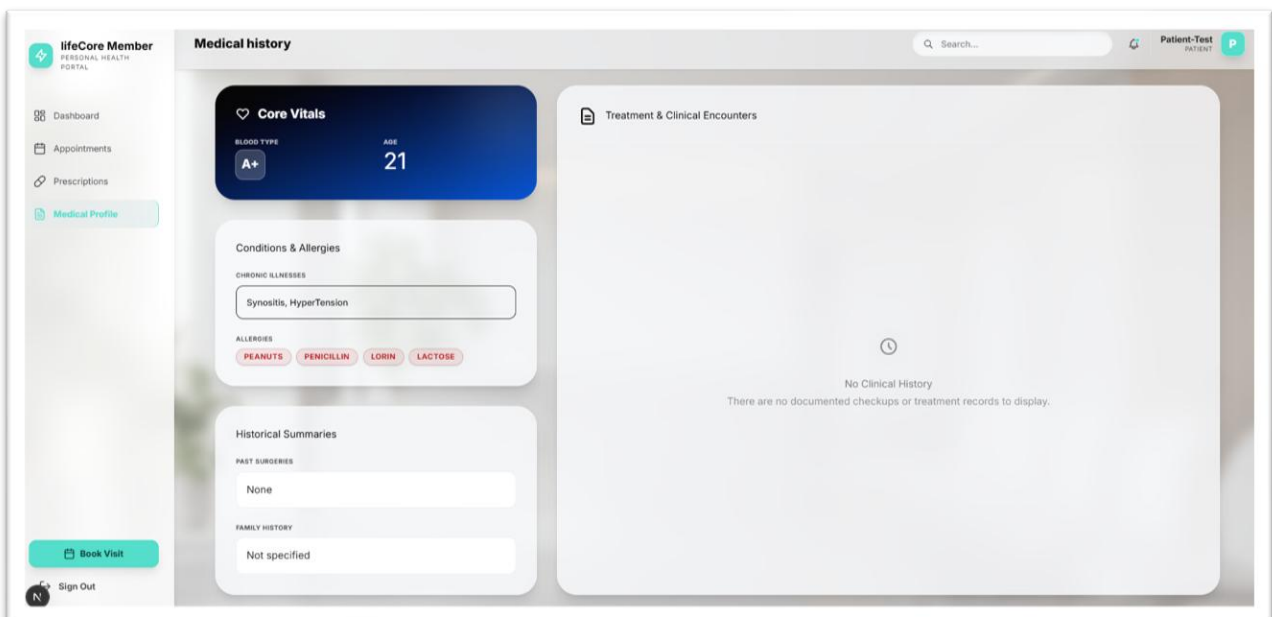


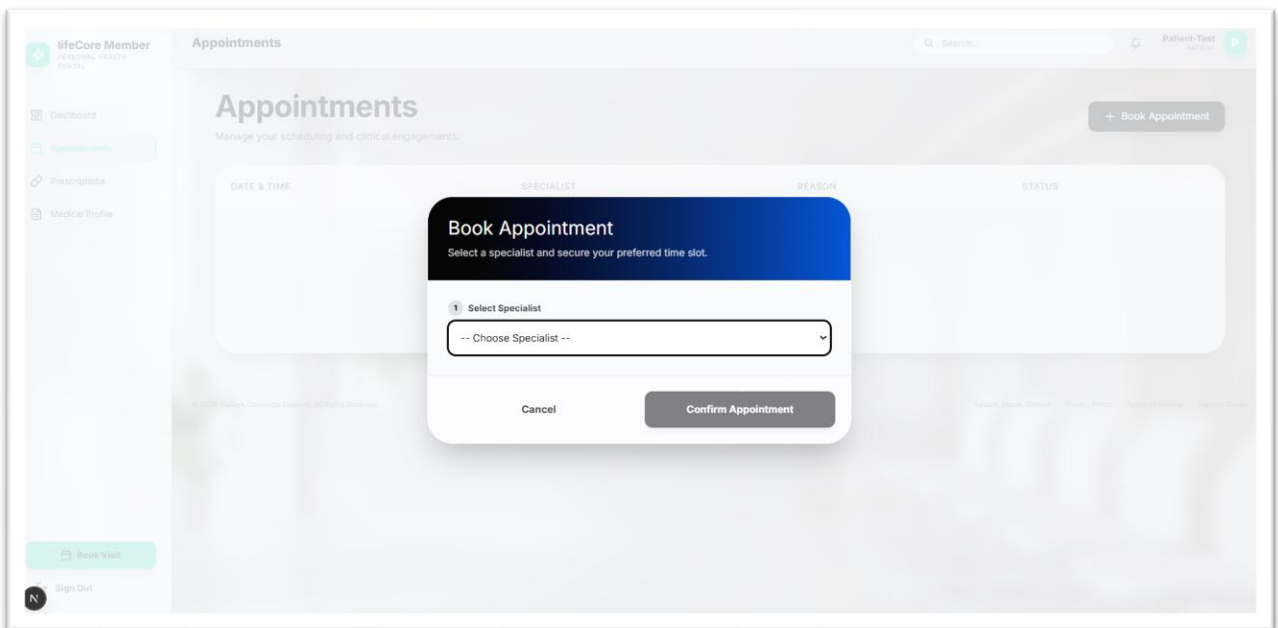
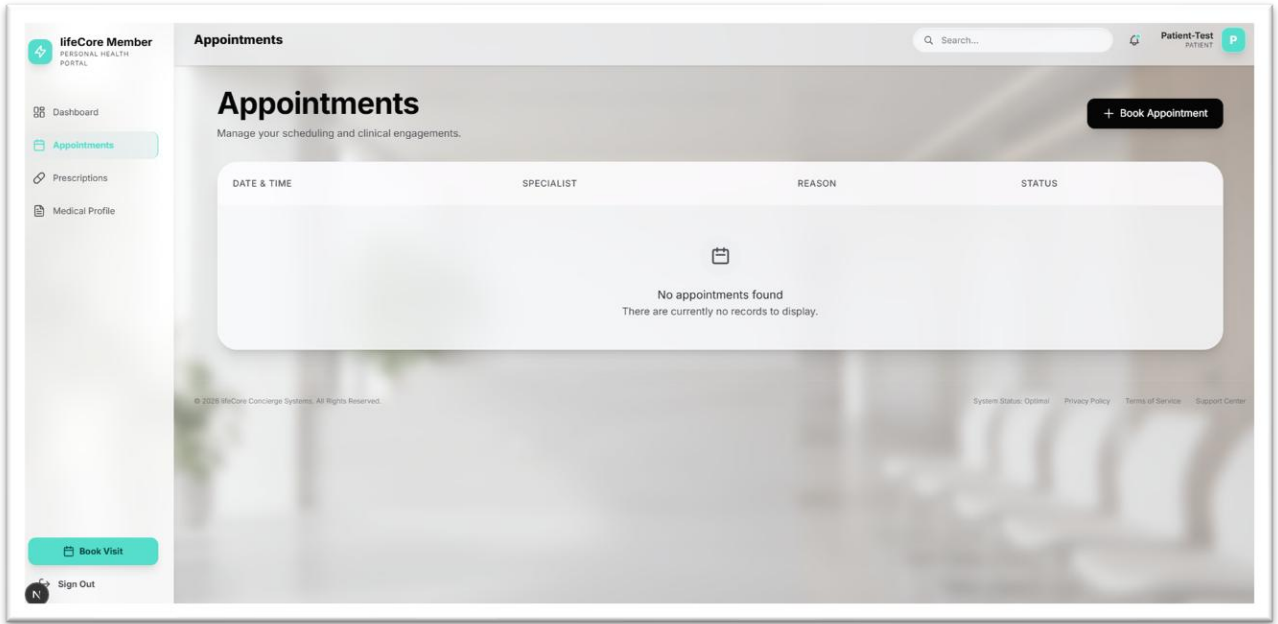
4. Module III: Appointment & Treatment Management System

This is the operational core of the hospital, facilitating real-time scheduling and detailed medical record tracking.

Implementation Details:

- **Appointment Booking:** Patients can request appointments based on their assigned doctor's specific daily availability. The system incorporates validation to prevent double-booking.
- **Status Workflow:** Appointments move dynamically through statuses (Pending, Confirmed, In Treatment, Completed, Rejected).
- **Automated Email Notifications:** Any change in appointment status automatically triggers a beautifully formatted HTML email sent to the patient's registered email address.
- **Prescriptions:** Doctors can seamlessly attach digital prescriptions to appointments, detailing medication names, dosages, frequencies, and durations. Patients are notified via email when a prescription is issued.
- **Treatment Tracking:** The platform tracks physical checkup metrics (height, weight, BP, Heart Rate) and treatment progress over time, storing medical history for easy retrieval.
-





Book Appointment

Select a specialist and secure your preferred time slot.

1 Select Specialist

Dr. Shamoil — Cardiology

2 Select Date

06/25/2026

3 Available Time Slots

09:00 AM

10:00 AM

11:00 AM

01:00 PM

02:00 PM

03:00 PM

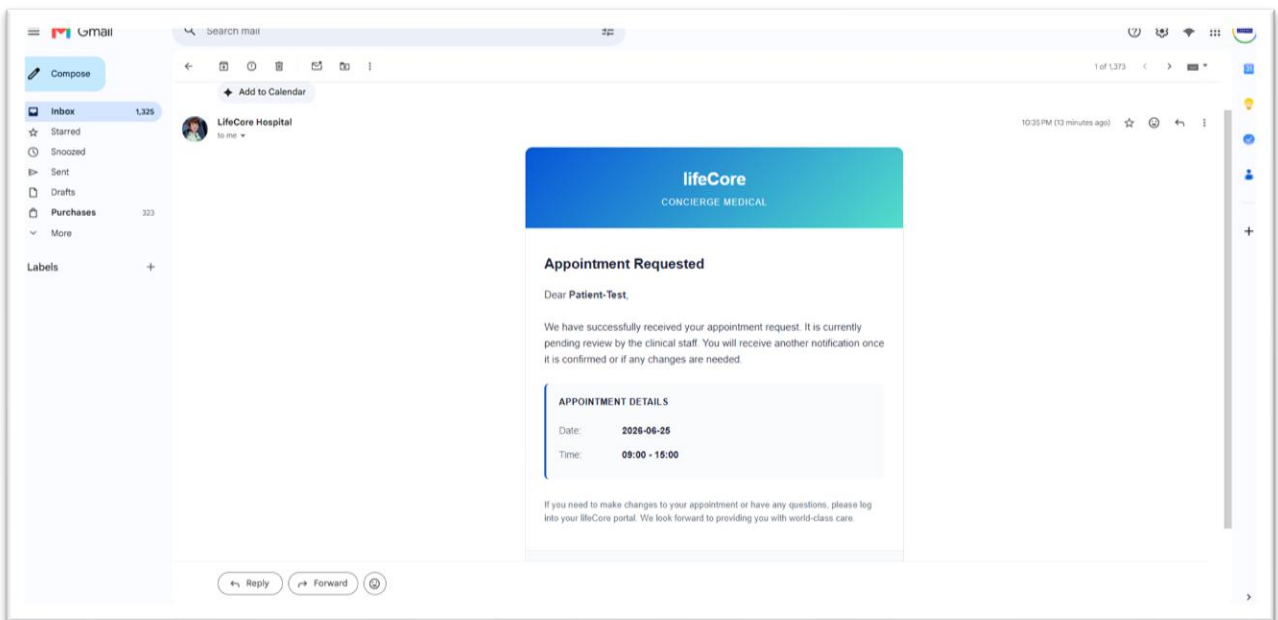
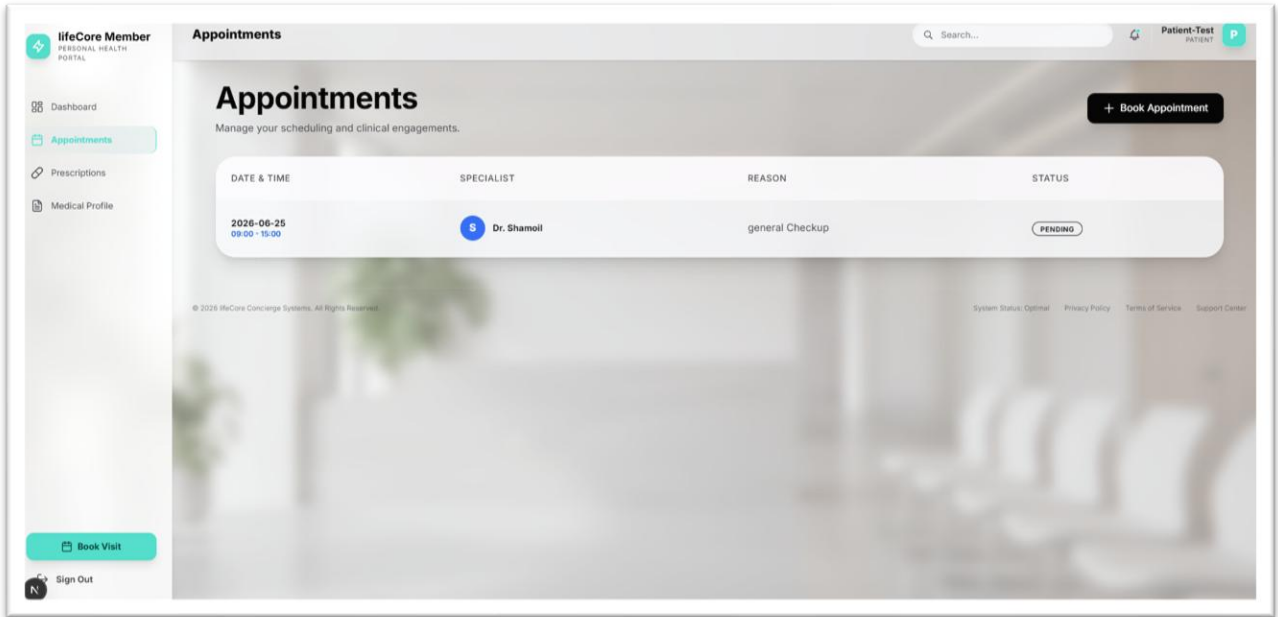
04:00 PM

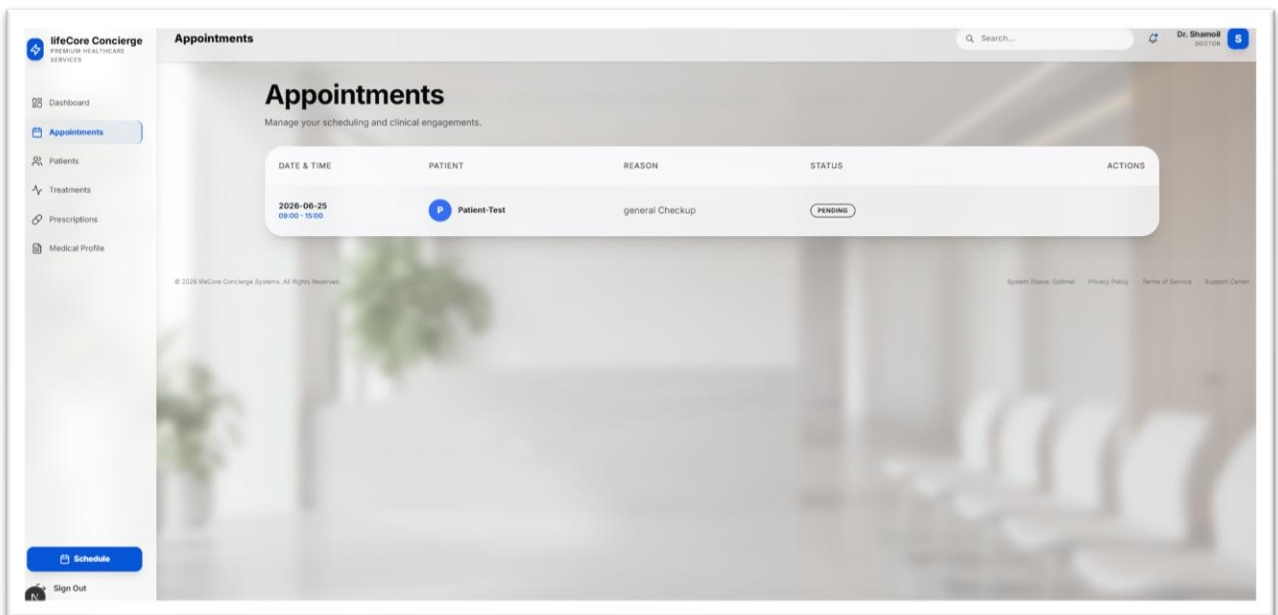
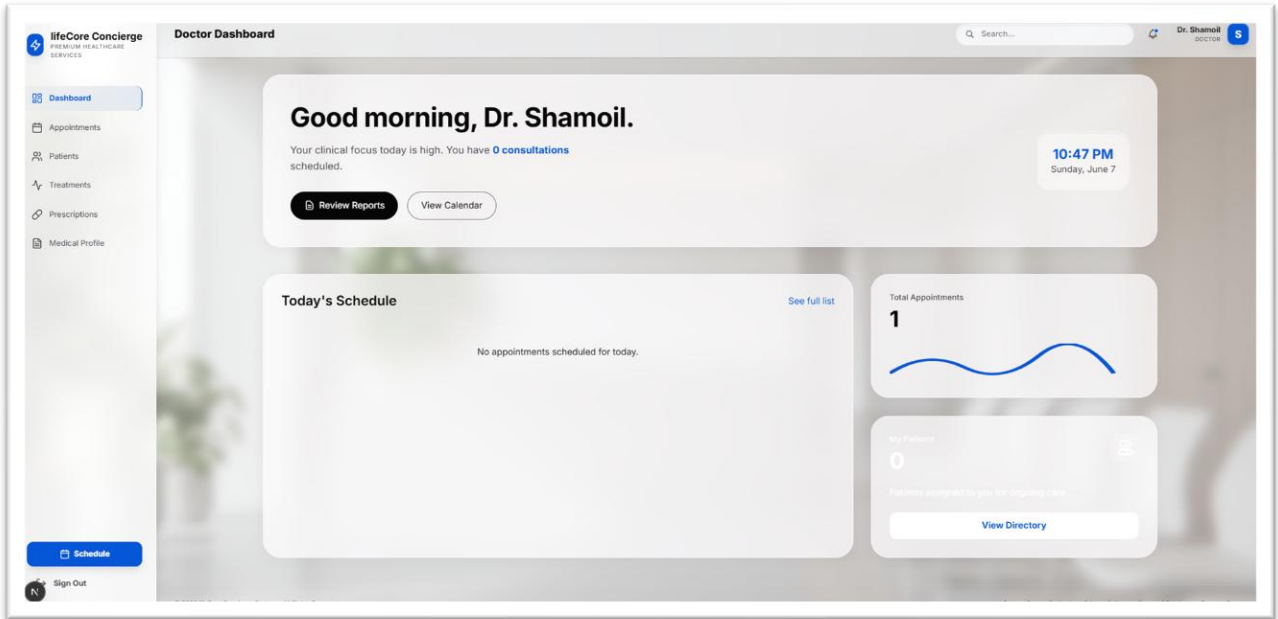
4 Reason for Visit

general Checkup

Cancel

Confirm Appointment





lifeCore Concierge
PREMIUM HEALTHCARE
SERVICES

Dashboard

Appointments

Patients

Treatments

Prescriptions

Medical Profile

Schedule

Sign Out

Appointments

Manage your scheduling and clinical engagements.

DATE & TIME	PATIENT	REASON	STATUS	ACTIONS
2026-06-25 09:00 - 15:00	P Patient-Test	general Checkup	CONFIRMED	

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System Status: Optimal | Privacy Policy | Terms of Service | Support Center

LifeCore Hospital

to me

10:48 PM (1 minute ago)

☆ 🗨 ↶

lifeCore
CONCIERGE MEDICAL

Appointment Confirmed

Dear Patient-Test,

Your appointment status has been updated to **Confirmed** by our clinical staff.

APPOINTMENT DETAILS

Date: 2026-06-25

Time: 09:00 - 15:00

If you need to make changes to your appointment or have any questions, please log into your lifeCore portal. We look forward to providing you with world-class care.

← **Treatment & Checkup**

Select Active Patient Appointment

Patient-Test - 2026-06-25 at 09:00 - 15:00 (Confirmed) ▼

Physical Checkup

Height (cm)

Weight (kg)

Blood Pressure

Heart Rate

Temperature

General Notes *

Record Checkup

Treatment Progress

Visit Type

Patient Status

Progress Notes *

Next Follow-up Date

Update Progress

5. Administrative Controls

The platform provides an overarching administrative view to oversee hospital operations, ensuring smooth management of both staff and patients. Admins possess the highest clearance for system-wide CRUD operations.

